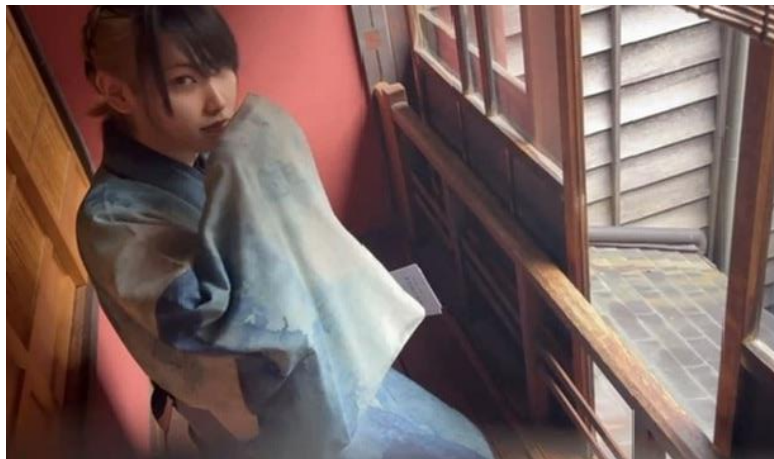


Hands-on Gate-based Quantum Computing with Qiskit & an Application

Taisei Matsuo (Ohzeki group)

Self Introduction



Name : Taisei Matsuo (he/his)
Post : Visitor researcher at uw
Research Interest :

- Tensor Network
- Random Quantum Circuit
- Generative Model
- etc.

Experience :

- “Tensor Renormalization Group Analysis of Phase Transition in Random tensor network model”
 1. Autumn Meeting of the Physical Society of Japan, Sep 2025.
 2. Tensor Network States: Algorithms and Applications (TNSAA) in Taiwan, Dec 2025.
- Organizer: Qiskit Fall Fest 2025 at Institute of Science Tokyo, Nov 2025

Hobby :

- Anime !
ex) Death note, NARUTO, Attack on Titan, Tokyo Ghoul...

Self Introduction

LinkedIn



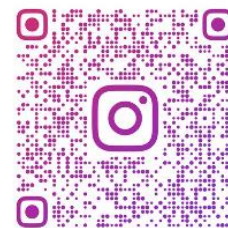
Twitter



github



Instagram



@INSTA_FOR_RENRAKU

Our Goal

Hands-on Section

How to use Quantum Circuit

- Install **Qiskit**
- Prepare some **Quantum State**
- Operate basic **Quantum Circuit**

30~40 min

Application Section

Implement a Quantum Algorithm

- Lecture for **Quantum Born Machine**
- Code the **Quantum Algorithm**
- Appendix : **Future works**

10~20 min

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What is Qiskit?

Python based open-source Software Development Kit developed by **IBM**

- Operate Virtual Quantum Circuits(QC)
- Implement Quantum Algorithms
- Connect with Real Quantum Device

Why use Qiskit?

- Can use QC in Real Hardware without major change
- Can simulate within practical noise
- Have sort of Library
 - 1.Optimization
 - 2.Machine Learning
 - 3.Chemistry

How to access Notebook

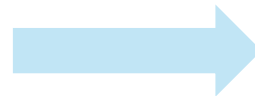
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4. Lastly, a fun fact / a heartfelt request. The organizing team has conducted extensive spectroscopic measurements on two fluids that are of paramount importance for scientists: coffee and water (in that order). Sadly, drinking coffee was still found not to count as drinking water. Thus, we encourage all participants to stay adequately hydrated. :)

Workshop website & program:

<https://altema.is.tohoku.ac.jp/expo/poland/en/>

Yesterday's mail



Programme

Overall Schedule

[Notebooks for Hands-On Sessions](#)

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“hands-on basic ideas of qiskit”

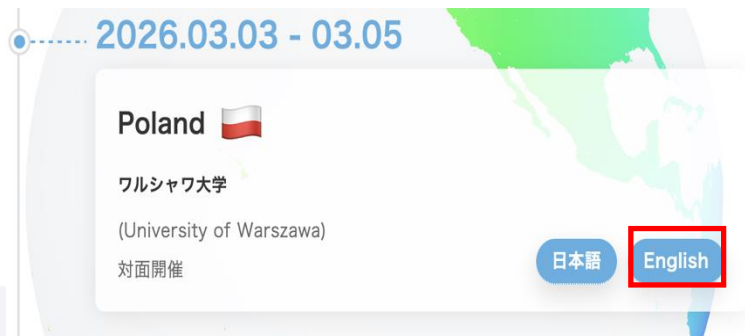
Intro



Tohoku University
https://aitema.is.tohoku.ac.jp/q_universe
Quantum Universe for You

1日前 — 大関 真之教授が東北大学「2025（令和7）年度総長教育賞」を受賞しました！総長教育賞は、東北大学の教育理念に基づき、誠意と熱意をもって職務に取り組まれ...

Search “quantum universe Tohoku univ”



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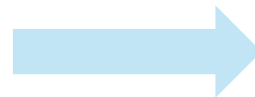
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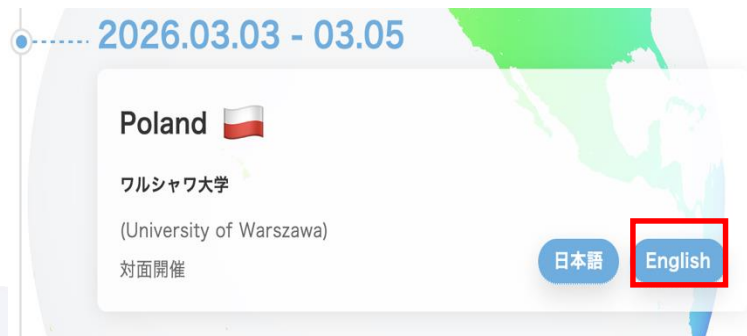
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Programme

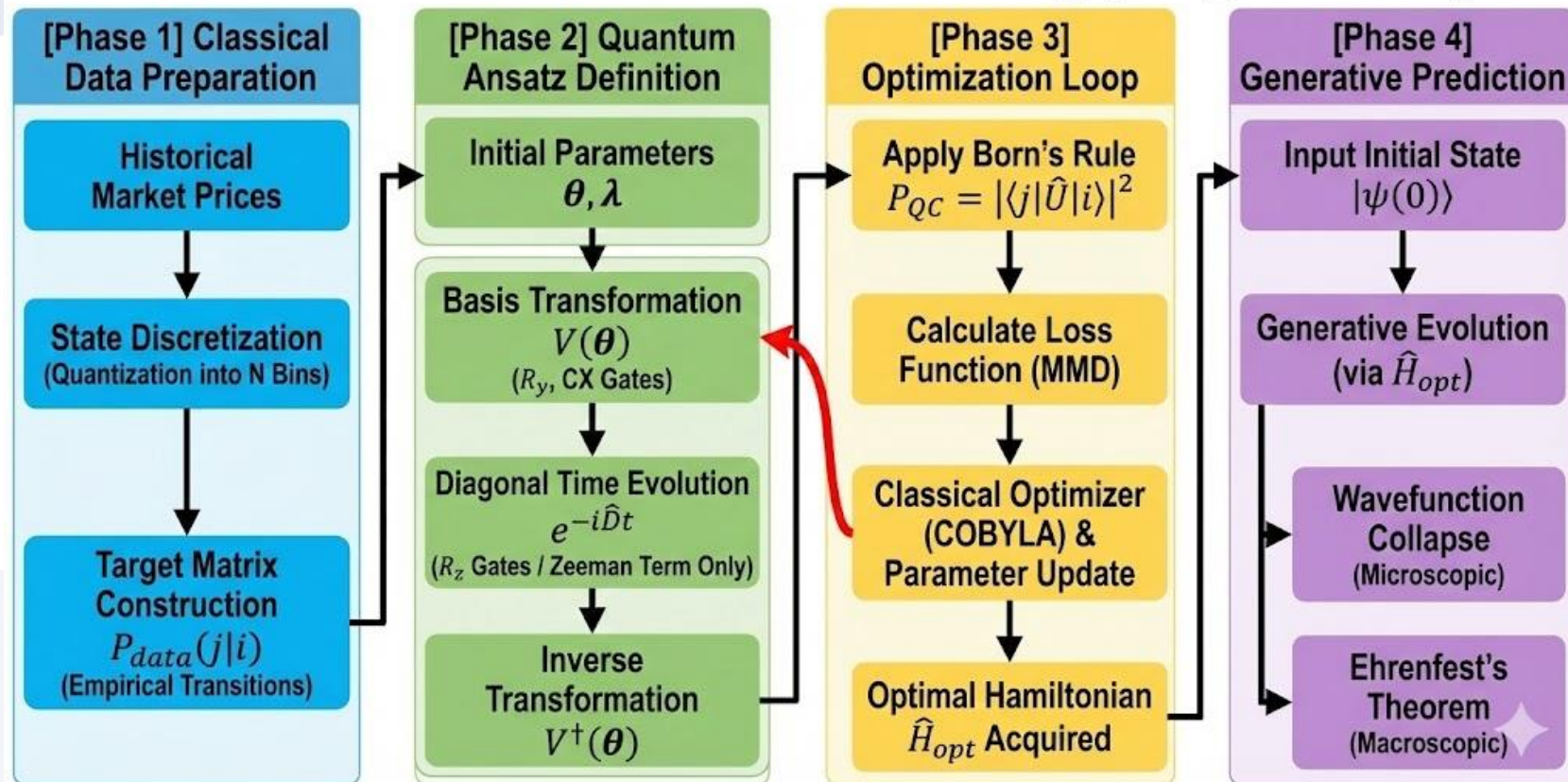
Overall Schedule

[Notebooks for Hands-On Sessions](#)

Open the Link

[“QI4U Quantum Born Machine.ipynb”](#)

Quantum-Native Financial Modelling (Single Asset)

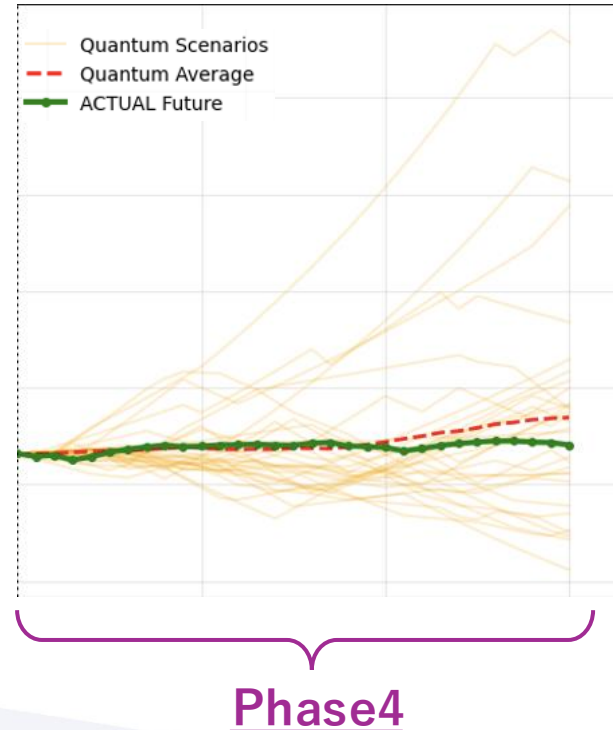


Our purpose

Predicting stock price dynamics over time.

- ➔ Mapping price volatility to Quantum States
- ➔ Learning the Hamiltonian via Quantum Circuits.
- ➔ Simulating future stock price movements.

➔ Earning money by using QC!



Probability distribution

Target transition matrix from data

$$\underbrace{P_{\text{data}}(j|i)}_{\text{Phase1}} = \frac{\text{Count}(|i\rangle \rightarrow |j\rangle)}{\sum_k \text{Count}(|i\rangle \rightarrow |k\rangle)}$$

Parameterized transition matrix

$$\underbrace{P_{\text{QC}}(j|i)}_{\text{Phase2}} = |\langle j|\hat{U}(\boldsymbol{\theta}, \boldsymbol{\lambda})|i\rangle|^2$$

Our Goal Phase3

Optimization problem of finding $\boldsymbol{\theta}$ and $\boldsymbol{\lambda}$ that minimize the loss function $\mathcal{L}$

$$\mathcal{L} = \sum_{i,j} (P_{\text{QC}}(j|i) - P_{\text{data}}(j|i))^2$$

Main

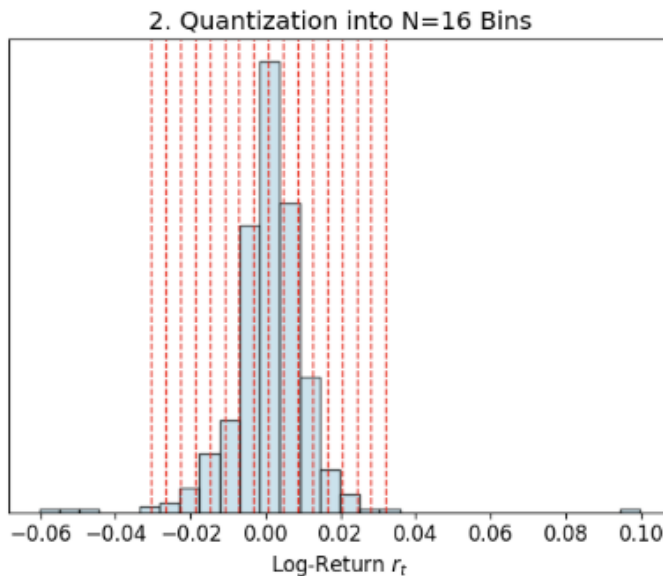
Phase1

Data

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right)$$

- S_t : raw asset price
- r_t : log return
- t : time series

Separated by **2⁴(4-qubit)**



Quantum States

$$|0000 = 0\rangle$$

$$|0001 = 1\rangle$$

⋮

$$|1111 = 15\rangle$$

$$P_{\text{data}}(j|i)$$

Main

Phase2

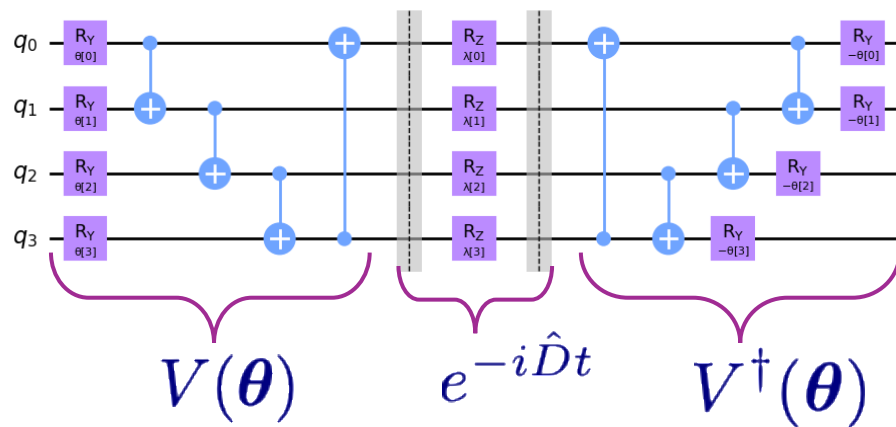
$$P_{QC}(j|i) = |\langle j | \hat{U}(\boldsymbol{\theta}, \boldsymbol{\lambda}) | i \rangle|^2$$

Unitary Operator

$$\begin{aligned}\hat{U}(t) &= e^{-i\hat{H}t} \\ &= V(\boldsymbol{\theta})e^{-i\hat{D}t}V^\dagger(\boldsymbol{\theta})\end{aligned}$$

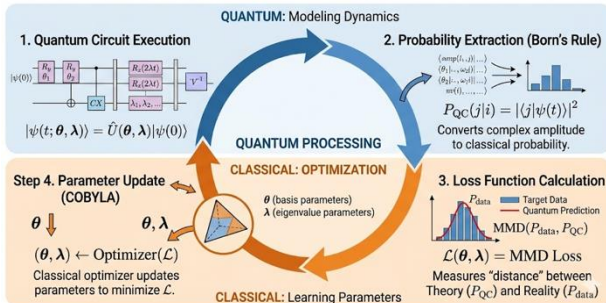
$$\begin{aligned}V(\boldsymbol{\theta}) &= \prod_{l=1}^L \left(U_{CX} \cdot \bigotimes_{q=0}^{n-1} R_y(\theta_{l,q}) \right) \\ e^{-i\hat{D}t} &= \bigotimes_{k=0}^{n-1} R_z(2\lambda_k t)\end{aligned}$$

Quantum Gate



$$\mathcal{L} = \sum_{i,j} (P_{\text{QC}}(j|i) - P_{\text{data}}(j|i))^2$$

Hybrid Quantum-Classical Optimization Loop for Hamiltonian Learning

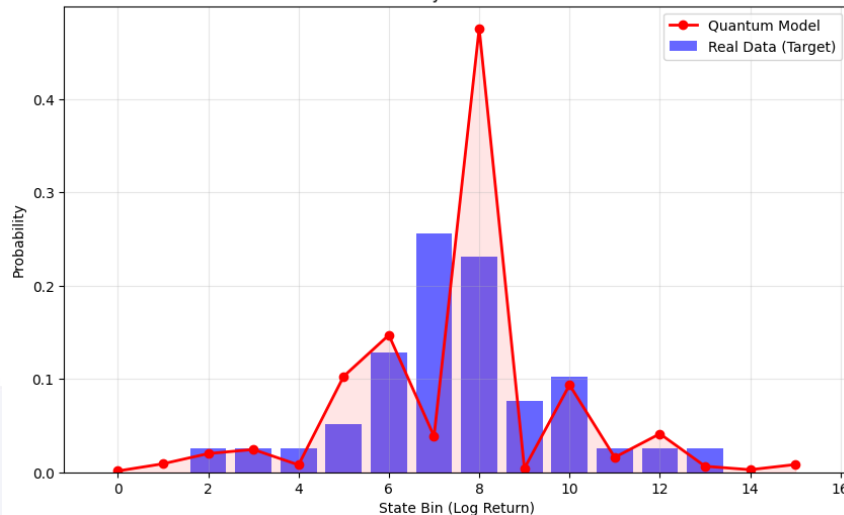


Time expansion operator

$$\theta_{\text{opt}}, \lambda_{\text{opt}} \rightarrow \hat{U}_{\text{opt}}(t)$$

Competition data to QC

Quantum Learning Performance
Fidelity = 78.23%



Main

Phase4

Future quantum state

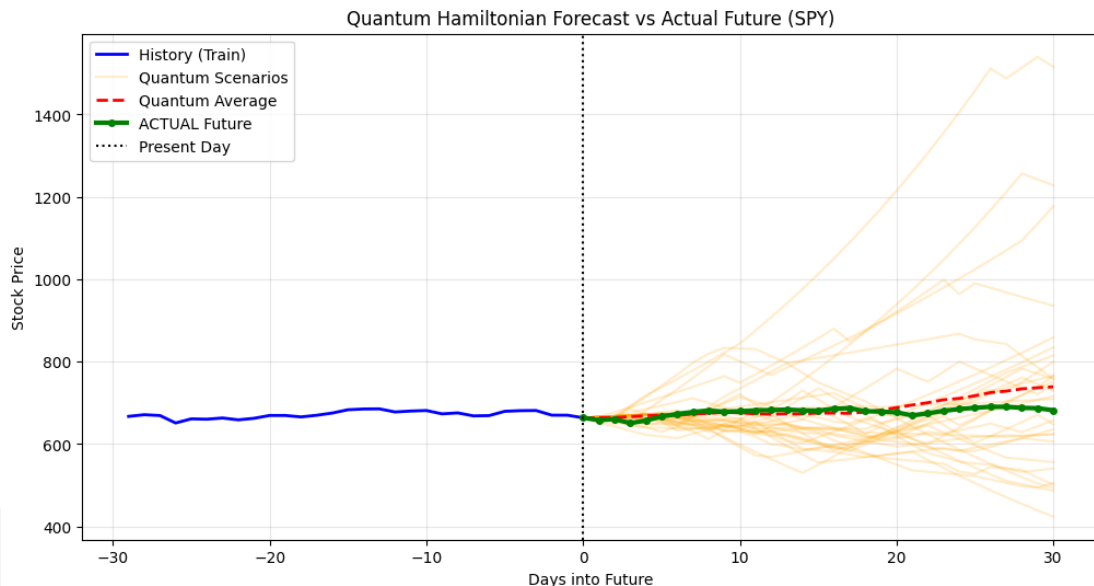
$$|\psi(t)\rangle = \hat{U}_{opt}(t)|\psi(0)\rangle$$

Future data(sampling)

$$r_t \sim P(j) = |\langle j|\psi(t)\rangle|^2$$

Future data(expectation)

$$\langle r_t \rangle = \langle \psi(t) | \hat{r} | \psi(t) \rangle$$



Conclusion

Classical ML

1. Lack of physical or financial meaning
(Neural Network)
2. Relies on unstable generative model
(GANs, VAEs)
3. Need other models for trend & risk
(LSTM, Monte Carlo)

Quantum ML

1. Learn eigenbasis & energy eigenvalue
(θ, λ)
2. Relies on Quantum Born Machine
(Entanglement)
3. Need only model for trend & risk
(Ehrenfest $\langle r_t \rangle, r_t$)

Not just predicting the data; Simulating the governing physical laws of markets